

Angela Chen

Vancouver, BC | angelachen37073@gmail.com | (604) 722 3601 | [Portfolio](#)

Education

University of British Columbia

Sept 2024 – May 2029

Engineering Physics (GPA 4.0/4.33)

- **Relevant Coursework** Systems and Controls, Signals and Systems, Software Construction, Experimental Techniques, Linear Circuits, Electricity and Magnetism, Mechanics, Applied Linear Algebra, ODEs, PDEs

Skills

Languages Python (NumPy, Flask, OpenCV), C/C++, Java, HTML, CSS, JS

Software & Tools Altium, KiCAD, Onshape, Git, Linux, MATLAB, LTspice

Laboratory & Hardware Soldering, Oscilloscope, Multimeter, Function Generator, Logic Analyzer, 3D Printing, Raspberry Pi, ESP32, UART, I2C

Experience

UBC Rocket

Sept 2026 – Present

Avionics Hardware Subteam Member

- Testing, validating and designing avionics hardware for the team's three rocket projects: competition (COTS) rocket, thrust-vector-controlled (TVR) rocket and liquid-propellant engine rocket.

UBC MINT (Multifaceted Innovation in Neurotechnology)

Jan 2025 – Present

Electrical Subteam Member

- Designed a 4-channel EMG analog front end for an EMG/EEG-controlled robotic arm, conditioning low-amplitude muscle signals for a 24-bit simultaneous-sampling SPI ADC using instrumentation amplifiers, two-stage $\sim 212\times$ gain, Sallen-Key anti-aliasing filters, and a right-leg-drive circuit to actively cancel 60 Hz common-mode noise.
- Developing the biosignal-acquisition hardware end to end: schematic capture, single-supply 3.3 V / 1.65 V virtual-ground power architecture, component selection, and 4-layer PCB layout in Altium.
- Designed a 4-layer single-cell lithium-ion battery charger PCB in Altium while successfully addressing engineering considerations in thermal management, power integrity and grounding strategies.

Centre for Accessibility, University of British Columbia

Jan 2026 – May 2026

Designated Notetaker, Part-time Job

- Enhanced accessibility of a technical course (PHYS 301: Electricity & Magnetism) for a student with a disability by producing clear, comprehensive notes across the full academic semester.

Projects

Autonomous Robot, UBC ENPH 253 Mars Rover Competition

May 2026 – August 2026

- Co-designed and built a fully autonomous, 4-wheel omni-drive robot with a 3-processor (dual ESP32 + Raspberry Pi) control architecture, integrating closed-loop encoder/PID motor control, IR-based line following, and a multi-claw system designed specifically for each competition task, as part of a 4-person team.
- Co-owned overall electrical architecture and 4 custom 2-layer PCBs including comparator-based IR line-following sensor boards, discrete MOSFET H-bridge motor driver board, a multi-rail power distribution board, and drivetrain control logic (combinational H-bridge gate driving via 74HC-series gates, sensor MUXing).
- Developed firmware for the tape-following sensor array, including driver code and line-position error calculation for closed-loop PID correction, and implemented portions of the robot's autonomous task sequencing logic.
- Built a real-time computer-vision system for an object detection task (Raspberry Pi 4 + ESP32): trained and validated a custom YOLOv26-nano object detector via transfer learning (~ 0.99 precision, 0.99 mAP@50), then optimized it for edge inference with an NCNN export.
- Engineered the surrounding embedded software: a multithreaded object detection state machine with multi-frame voting for robustness, a framed binary UART protocol (checksum + parser state machine) to interface with the dual ESP32, and reliable boot-on-power deployment as a Linux systemd service.

Custom FPV Drone, Personal Project

September 2026 – Present

- Designing a custom FPV quadcopter with a STM32 flight controller and GPS in a 4-person group.

Awards

2026 UBC ENPH 253 Autonomous Robot Competition 3rd Place (out of 16 teams), Dean's List | 2025 Dean's List | 2024 District Authority Scholarship (Community Service), BC Achievement Scholarship, Lions Club Scholarship